Bayesian Estimation: Questions 1-10

Beta-Binomial and Gamma-Poisson conjugacy, loss functions, and Bayes estimators. Correct choices are marked ✓ in green.

Question 1 · Multiple Answer

You have independent Bernoulli(p) data, $X_1, \ldots, X_n$. The prior for p is Beta(α, β). Which statements below are **true**? (Select all that apply)

A. The posterior variance is always zero if α + β is very large

✓ **B. The posterior mean is $(\alpha + \Sigma X_i) / [(\alpha + \Sigma X_i) + (\beta + n - \Sigma X_i)]$**

C. The posterior is Beta(α + n, β) regardless of the data

✓ **D. If α = 1 and β = 1, the prior is uniform on [0, 1]**

Answer: Statements 2 and 4

The Beta prior is conjugate to the Bernoulli likelihood, so the posterior is Beta($\alpha + \Sigma X_i$, $\beta + n - \Sigma X_i$): add successes to α and failures to β.

1 is false. A large α + β makes the posterior variance small but never exactly zero. Var of Beta(a, b) = $ab / [(a+b)^2(a+b+1)] > 0$.

2 is true. The mean of Beta(a, b) is a/(a+b), which gives exactly this expression, simplifying to $(\alpha + \Sigma X_i) / (\alpha + \beta + n)$.

3 is false. That only happens if every observation is a success ($\Sigma X_i = n$).

4 is true. Beta(1, 1) has density proportional to $p^0(1 - p)^0 = 1$ on [0, 1].

Question 2 · Multiple Choice

A random sample of n = 12 items is tested for defectives. Let $X_i = 1$ if the i-th item is defective and 0 otherwise, and suppose $\Sigma X_i = 5$. The prior for θ, the proportion of defectives, is Beta(2, 3). What is the **posterior mean** for θ?

✓ **A. 7/17**

B. 7/15

C. 12/17

D. 2/5

Answer: 7/17

Posterior is Beta(2 + 5, 3 + 12 − 5) = Beta(7, 10), with mean 7/(7 + 10) = 7/17.

Distractors: 2/5 is the prior mean, 7/15 leaves failures out of β, and 12/17 uses n in place of the defect count.

Question 3 · Multiple Choice

Which of the following statements about a loss function $L(\theta, a)$ is **incorrect**?

✓ **A. A loss function must be based on the squared difference $(\theta - a)^2$ to ensure non-negativity**

B. It measures how bad it is to estimate θ using a

C. If we knew θ, we could choose $a = \theta$ to suffer zero loss

D. The expected prior loss is $\int L(\theta, a)\, \pi(\theta)\, d\theta$

Answer: The first statement (it is the incorrect one)

Squared error is just one choice. Absolute error $|\theta - a|$, 0-1 loss, and asymmetric losses are all non-negative too. The Bayes estimator under absolute error is the posterior median.

The other three are correct descriptions of a loss function.

Question 4 · Multiple Choice

A website typically sees λ clicks per hour. We place a Gamma(3, 1) prior on λ. Over a 2-hour period, we observe $X_1 = 5$ clicks in the first hour and $X_2 = 7$ in the second hour (total 12 clicks). What is the **Bayes estimate** of λ under **squared error loss**?

A. 3

✓ **B. 5**

C. 6

D. 4.67

Answer: 5

Under squared error loss the Bayes estimate is the posterior mean. Gamma is conjugate to Poisson: posterior is $\text{Gamma}(\alpha + \Sigma X_i, \beta + n) = \text{Gamma}(3 + 12, 1 + 2) = \text{Gamma}(15, 3)$.

Posterior mean = 15/3 = 5. Compare with the prior mean (3) and sample mean/MLE (6): the estimate sits between them.

Question 5 · Multiple Choice

You are estimating the one-year return on Apple stock for options trading. Small errors are fine, but large errors could lose a lot of money, so you want to penalize big mistakes heavily. Choosing between squared error loss and absolute error loss, which is more appropriate?

A. Absolute error loss, because it penalizes outliers more severely.

B. Both penalize outliers equally.

✓ **C. Squared error loss, because it penalizes larger deviations quadratically.**

D. Neither loss function has to do with deviations from the true return.

Answer: Squared error loss

Squared loss grows with the square of the miss; absolute loss grows linearly. A miss of 0.5 costs 0.25 vs 0.5, but a miss of 10 costs 100 vs 10. Small errors are treated gently and big ones harshly, matching the scenario.

Absolute error is actually the more outlier-robust loss (its Bayes estimator is the posterior median).

Question 6 · Multiple Choice

Which of the following is correct?

A. The posterior is proportional to the prior times the marginal distribution of the data

✓ **B. The posterior is proportional to the likelihood times the prior**

C. The marginal distribution of the data is proportional to the posterior times the likelihood

D. The prior is proportional to the posterior times the likelihood

Answer: Posterior ∝ likelihood × prior

Bayes' theorem: $\pi(\theta \mid x) = f(x \mid \theta)\, \pi(\theta) / m(x)$, where $m(x) = \int f(x \mid \theta)\, \pi(\theta)\, d\theta$.

Since $m(x)$ does not depend on θ, it is a normalizing constant, so $\pi(\theta \mid x) \propto f(x \mid \theta)\, \pi(\theta)$.

Question 7 · Multiple Choice

In a Normal(μ, σ^2) model with **known** variance and a Normal(μ_0, ${\sigma_0}^2$) prior on μ, what happens if the prior variance ${\sigma_0}^2$ is very large?

A. The posterior mean is pulled strongly toward μ_0

B. The posterior variance is large, indicating high certainty

C. The marginal distribution of the data cannot be computed

✓ **D. The posterior mean is close to the sample mean**

Answer: The posterior mean is close to the sample mean

Posterior mean = $[(1/{\sigma_0}^2)\mu_0 + (n/\sigma^2)\bar{x}] / [(1/{\sigma_0}^2) + (n/\sigma^2)]$.

As ${\sigma_0}^2$ grows, the prior precision goes to 0, so the weight on μ_0 vanishes and the posterior mean approaches $\bar{x}$. A vague prior lets the data dominate.

Strong pull toward μ_0 happens when ${\sigma_0}^2$ is small; large variance means low certainty; the marginal is still Normal and computable.

Question 8 · Multiple Choice

A quality-control engineer inspects n = 50 electronic components and finds k = 7 defective. The unknown defect probability is p. The **prior** on p is Beta(2, 2). Under **squared error loss**, what is the Bayes estimate of p?

A. 8/52

B. 1/2

C. 7/50

✓ **D. 9/54**

Answer: 9/54 (= 1/6 ≈ 0.167)

Bayes estimate under squared error loss is the posterior mean. Posterior is Beta(2 + 7, 2 + 50 − 7) = Beta(9, 45), with mean 9/54.

Distractors: 1/2 is the prior mean, 7/50 is the MLE, and 8/52 corresponds to a Beta(1, 1) prior.

Question 9 · Multiple Choice

A Bayes estimator under a loss function is defined to be the value that:

✓ **A. Minimizes the expected posterior loss**

B. Minimizes the posterior distribution

C. Maximizes the prior distribution

D. Maximizes the likelihood

Answer: Minimizes the expected posterior loss

The Bayes estimator is the action a minimizing $E[L(\theta, a) \mid x] = \int L(\theta, a)\, \pi(\theta \mid x)\, d\theta$.

This gives the posterior mean under squared error and the posterior median under absolute error. Maximizing the likelihood gives the MLE instead.

Question 10 · Multiple Choice

A random sample of n = 12 items is tested for defectives, with $\Sigma X_i = 5$. The prior for θ, the proportion of defectives, is Beta(2, 3). What is the **prior mean** for θ?

✓ **A. 2/5**

B. 3/5

C. 1/25

D. 2/3

Answer: 2/5

The prior mean depends only on the prior, so the data do not matter. For Beta(α, β), the mean is α/(α + β) = 2/5.

3/5 is the prior mean of 1 − θ, and 2/3 is α/β, a common mix-up.